

ASSIGNMENT -11

ADVANCED PROGRAMMING

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```
1 from abc import ABC, abstractmethod
2
3
4 class LibraryItem(ABC):
5     total_items = 0 # static/class variable
6
7     def __init__(self, title, year):
8         self.title = title
9         self.year = year
10        LibraryItem.total_items += 1 # increment counter
11
12        @abstractmethod
13        def display_info(self):
14            pass
15
16        @classmethod
17        def get_total_items(cls):
18            return cls.total_items
19
20
21 class Book(LibraryItem):
22     def __init__(self, title, year, author):
23         super().__init__(title, year)
24         self.author = author
25
26     def display_info(self):
27         print(f"[Book] Title: {self.title}, Author: {self.author}, Year: {self.year}")
28
29
30 class DVD(LibraryItem):
31     def __init__(self, title, year, duration, genre):
32         super().__init__(title, year)
33         self.duration = duration
34         self.genre = genre
35
36     def display_info(self):
```

```

class DVD(LibraryItem):
    def __init__(self, title, year, duration, genre):
        self.duration = duration
        self.genre = genre

    def display_info(self):
        print(f"[DVD] Title: {self.title}, Genre: {self.genre}, Duration: {self.duration} mins, Year: {self.year}")

items = [
    Book("Clean Code", 2008, "Robert C. Martin"),
    DVD("Inception", 2010, 148, "Sci-Fi"),
    Book("The Pragmatic Programmer", 1999, "Andrew Hunt")

Polymorphism: same method call, different behavior
for item in items:
    item.display_info()

print(f"\nTotal items in library: {LibraryItem.get_total_items()}")

```

OUTPUT:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SPELL CHECKER

c:\Users\VITU\OneDrive\Desktop\lib2>python -u "c:\Users\VITU\OneDrive\Desktop\lib2\LibrarySystem.py"
[Book] Title: Clean Code, Author: Robert C. Martin, Year: 2008
[DVD] Title: Inception, Genre: Sci-Fi, Duration: 148 mins, Year: 2010
[Book] Title: The Pragmatic Programmer, Author: Andrew Hunt, Year: 1999

Total items in library: 3

c:\Users\VITU\OneDrive\Desktop\lib2>

```